

Eigenvalues and Eigenvectors

Equation : $Av = \lambda v$

Points to remember are

- A = square matrix
- $v \neq 0$ (eigenvector)
- λ = eigenvalue

Normally, multiplying a vector by a matrix changes

- its direction
- its length

But for some special vectors

- It only stretches, shrinks, or flips the vector.
- Those special vectors are eigenvectors.
- The amount of stretching is the eigenvalue.

Example:

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \quad v = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad Av = \begin{bmatrix} 3 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$(\lambda, v) = \left(3, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$$

How to Find Eigenpairs

Step 1: $\det(A - \lambda I) = 0$

Step 2 : $(A - \lambda I)v = 0$

Example of finding:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\det \left(\begin{bmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{bmatrix} \right) = 0$$

$$(4 - \lambda)(3 - \lambda) - 2 = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 5)(\lambda - 2) = 0$$

$$\boxed{\lambda = 5, 2}$$

Finding with Eigenvalue 2

$$(A - 2I) = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$$2x + y = 0$$

$$x = 1$$

$$y = -2$$

Then eigenvector will be $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

finding with Eigenvalue 5

$$(A - 5I)v = 0$$

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} v = 0$$

$$-x + y = 0$$

$$y = x$$

$$v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Eigenpairs are

$$\boxed{\left(5, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)}$$

$$\boxed{\left(2, \begin{bmatrix} 1 \\ -2 \end{bmatrix} \right)}$$

Term	Meaning
Eigenvalue	Scaling factor λ
Eigenvector	Non-zero vector whose direction is preserved
Eigenpair	(λ, v) satisfying ($Av = \lambda v$)
Characteristic Equation	$\det(A - \lambda I) = 0$
Eigenvector Equation	$(A - \lambda I)v = 0$

Diagonalization of a Matrix

1. What is Diagonalization?

A square matrix A is diagonalizable if it can be written as

$$A = S\Lambda S^{-1}$$

Or

$$S^{-1}AS = \Lambda$$

where

- S = matrix of eigenvectors
- Λ = diagonal matrix of eigenvalues
- S^{-1} = inverse of S

2. Why Diagonalize?

A diagonal matrix is much easier to work with.

Instead of computing powers of

$$A^k$$

We compute

$$\Lambda^k$$

3) Structure of S and Λ

Suppose A has n eigenpairs

$$(\lambda_1, v_1), (\lambda_2, v_2), \dots, (\lambda_n, v_n)$$

Then S is the matrix of eigenvectors as columns

$$S = \begin{bmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{bmatrix}$$

Diagonal matrix of Eigenvalues

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Important: The order of eigenvalues in Λ must match the order of the corresponding eigenvectors in S.

4. Why Does It Work?

Since

$$Av_i = \lambda_i v_i$$

$$AS = [Av_1 \quad Av_2 \quad \cdots \quad Av_n] = [\lambda_1 v_1 \quad \lambda_2 v_2 \quad \cdots \quad \lambda_n v_n]$$

$$AS = S\Lambda$$

$$A = S\Lambda S^{-1}$$

5. When is a Matrix Diagonalizable?

An $n \times n$ matrix is diagonalizable if and only if it has

n linearly independent eigenvectors.

This is the most important condition.

6. Easy Test

If a matrix has

n distinct eigenvalues,

then it is automatically diagonalizable.

Reason: Distinct eigenvalues always have linearly independent eigenvectors.

Example

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 1 \end{bmatrix}$$

Eigenvalues = 5 and -1

Eigenvectors are

$$v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -\frac{5}{2} \end{bmatrix}$$

$$S = \begin{bmatrix} 2 & 1 \\ 1 & -\frac{5}{2} \end{bmatrix} \quad \Lambda = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A = S\Lambda S^{-1}$$

Computing Matrix Powers

$$A = S\Lambda S^{-1}$$

$$A^2 = (S\Lambda S^{-1})(S\Lambda S^{-1}) = S\Lambda^2 S^{-1}$$

$$\boxed{A^k = S\Lambda^k S^{-1}.}$$

Power of a Diagonal Matrix

$$\Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \quad \Lambda^3 = \begin{bmatrix} 2^3 & 0 \\ 0 & 5^3 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 0 & 125 \end{bmatrix}$$

Quick Revision Table

Concept	Formula / Idea
Diagonalization	$A = S\Lambda S^{-1}$
Equivalent form	$S^{-1}AS = \Lambda$
Columns of S	Eigenvectors
Diagonal of Λ	Eigenvalues
Condition	n linearly independent eigenvectors
Easy Test	n distinct eigenvalues $\Rightarrow$ diagonalizable
Matrix powers	$A^k = S\Lambda^k S^{-1}$
Λ^k	Raise each diagonal entry to the k-th power
Uniqueness	S is not unique; Λ 's order depends on the ordering of eigenvectors

Most Important Formulas

$$A = S\Lambda S^{-1}$$

$$S^{-1}AS = \Lambda$$

$$AS = S\Lambda$$

$$A^k = S\Lambda^k S^{-1}$$

Distinct eigenvalues $\Rightarrow$ Diagonalizable

Example for Diagonalization of a Matrix

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$$

We will find

- Eigenvalues
- Eigenvectors
- S
- Λ
- S^{-1}
- Verify $A = S\Lambda S^{-1}$

Step 1 : Find the Eigen-Values

$$\det(A - \lambda I) = 0$$

$$A - \lambda I = \begin{bmatrix} 4 - \lambda & 2 \\ 1 & 3 - \lambda \end{bmatrix}$$

$$\lambda_1 = 5, \quad \lambda_2 = 2$$

$$(4 - \lambda)(3 - \lambda) - 2 = 0$$

$$12 - 4\lambda - 3\lambda + \lambda^2 - 2 = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

Step-2: Find the Eigenvectors for eigenvalue 5

$$v = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$(A - 5I)v = 0$$

$$A - 5I = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}$$

$$-x + 2y = 0$$

$$x = 2y$$

$$v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Choose $y = 1$

Then $x = 2$

Step-3: Find the Eigenvector for Eigenvalue 2

$$(A - 2I)v = 0$$

$$A - 2I = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$$

$$2x + 2y = 0$$

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Or

$$x + y = 0$$

Choose $x = 1$, then $y = -1$

Step-4: Construct the matrix of Eigenvectors and diagonal matrix of eigenvalues

$$S = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \Lambda = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

First column $\leftrightarrow$ first eigenvalue

Second column $\leftrightarrow$ second eigenvalue

Always keep this order.

Step-5: Find the Inverse of S matrix

Append the identity matrix to the right.

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{array} \right] \quad S^{-1} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{bmatrix}$$

The goal is to transform the left side into the identity matrix.

Step-6: Verification

$$S \Lambda S^{-1} = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} = A$$

Orthogonality diagonalizable matrices

1. Real Symmetric Matrix

A matrix is real symmetric if

$$A = A^T$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \quad A^T = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = A$$

2. Spectral Theorem

The Spectral Theorem states:

Every real symmetric matrix can be orthogonally diagonalized.

That means there always exists an orthogonal matrix Q such that

$$A = Q \Lambda Q^T \quad Q^T A Q = \Lambda.$$

3. What Does the Spectral Theorem Guarantee?

If A is real and symmetric, then:

- Every eigenvalue is real. No imaginary eigenvalues.
- Eigenvectors for different eigenvalues are orthogonal.

$$\lambda_i \neq \lambda_j \quad \boxed{v_i^T v_j = 0.}$$

→ We can choose all eigenvectors to be orthonormal.

Meaning

Each vector has length 1.

$$\|v_i\| = 1$$

→ Every real symmetric matrix is diagonalizable. No additional test is needed.

4. Orthogonal Matrix

A matrix

$$Q = [v_1 \ v_2 \ \cdots \ v_n]$$

is orthogonal if

$$\boxed{Q^T Q = Q Q^T = I.}$$

This is the biggest advantage over ordinary diagonalization.

5. Ordinary vs Orthogonal Diagonalization

Ordinary Diagonalization	Orthogonal Diagonalization
$A = S \Lambda S^{-1}$	$A = Q \Lambda Q^T$
S^{-1} must be computed	$Q^{-1} = Q^T$
Works only if enough independent eigenvectors exist	Works for every real symmetric matrix

Columns of (S): independent eigenvectors	Columns of (Q): orthonormal eigenvectors
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6. How to Orthogonally Diagonalize

Given a symmetric matrix:

Step 1: Find eigenvalues.

$$\det(A - \lambda I) = 0$$

Step 2: Find eigenvectors.

$$(A - \lambda I)v = 0$$

Step 3: Normalize every eigenvector.

$$v = \begin{bmatrix} a \\ b \end{bmatrix} \quad u = \frac{v}{\|v\|} \quad \|v\| = \sqrt{a^2 + b^2}.$$

Now every vector has length 1.

Step 4 : Construct

$$Q = [u_1 \ u_2 \ \cdots] \quad \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots).$$

Step 6: Write

$$A = Q\Lambda Q^T.$$

Complete Example

Diagonalize

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Step 1: Find eigenvalues.

$$\det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = 0 \quad \begin{aligned} (2 - \lambda)^2 - 1 &= 0 \\ \lambda^2 - 4\lambda + 3 &= 0 \\ (\lambda - 3)(\lambda - 1) &= 0 \end{aligned}$$

$$\lambda_1 = 3, \quad \lambda_2 = 1.$$

Step 2 Eigenvector for eigenvalue = 3

$$(A - 3I)v = 0. \quad -x + y = 0.$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Step 3 : Eigenvector for eigenvalue = 1

$$A - I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad x + y = 0.$$

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$v_1^T v_2 = 1 - 1 = 0$$

Notice that the column vectors are already orthogonal to each other.

Step 4 : Normalize

$$\|v_1\| = \sqrt{2}. \quad \|v_2\| = \sqrt{2}.$$

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Step 5 : Construct other required matrices

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

Step 7 : Write the answer

$$A = Q\Lambda Q^T.$$

Since, there is **no need to compute the inverse**.

$$Q^{-1} = Q^T$$

Concept	Formula / Idea
Symmetric matrix	$A = A^T$
Spectral Theorem	Every real symmetric matrix is orthogonally diagonalizable
Orthogonal matrix	$Q^T Q = I$
Inverse	$Q^{-1} = Q^T$
Orthogonal diagonalization	$A = Q\Lambda Q^T$
Columns of (Q)	Orthonormal eigenvectors

Diagonal of Λ	Eigenvalues
Distinct eigenvectors	Automatically orthogonal

Complex Matrices and Hermitian Matrices

Complex Numbers

A complex number has the form

$$z = a + ib$$

where

- a = real part
- b = imaginary part
- $i = \sqrt{-1}$

$$3 + 2i, \quad -1 + 4i, \quad 5 - i$$

Complex Conjugate

The conjugate changes the sign of the imaginary part.

$$(a + ib)^* = a - ib$$

Number	Conjugate
(3+2i)	(3-2i)
(5-i)	(5+i)
(7)	(7)

Complex Inner Product

Instead, define

$$\langle x, y \rangle = x^* y$$

where x^* is the conjugate transpose.

This guarantees

$$x^* x \geq 0.$$

Conjugate Transpose

The conjugate transpose consists of two steps:

Step 1 : Transpose the matrix.

Step 2 : Take the complex conjugate of every entry.

Notation A^*

Read as "A star" or "Hermitian transpose."

Example for Conjugate Transpose of a Complex Matrix

$$A = \begin{bmatrix} 1+i & 2 \\ 3-i & 4i \end{bmatrix} A^T = \begin{bmatrix} 1+i & 3-i \\ 2 & 4i \end{bmatrix} A^* = \begin{bmatrix} 1-i & 3+i \\ 2 & -4i \end{bmatrix}$$

Hermitian Matrix

A matrix is Hermitian if

$$A^* = A.$$

This is the complex equivalent of a symmetric matrix.

Example of Hermitian Matrix

$$A = \begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix} \begin{bmatrix} 2 & 3-i \\ 3+i & 5 \end{bmatrix} \begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix} = A$$

Transpose the
Matrix A

Conjugate

Therefore A is a Hermitian Matrix.

Real Matrix	Complex Matrix
Symmetric	Hermitian
$A = A^T$	$A = A^*$
Entries are real	Entries may be complex
Eigenvalues are real	Eigenvalues are also real

Properties of a Hermitian Matrix

A Hermitian matrix satisfies:

- (a) It is equal to its conjugate transpose.
- (b) Diagonal entries are always real.

What is a Unitary Matrix?

A matrix U is unitary if

$$U^*U = UU^* = I.$$

$$U^{-1} = U^*.$$

Real case: $Q^{-1} = Q^T$

Complex Case : $U^{-1} = U^*$

What Does "Unitary" Mean?

The columns of a unitary matrix are

- mutually orthogonal
- each has length 1

Hence the columns are orthonormal.

Real Space	Complex Space
Symmetric	Hermitian
$A = A^T$	$A = A^*$
Orthogonal matrix (Q)	Unitary matrix (U)
$Q^{-1} = Q^T$	$U^{-1} = U^*$
$A = Q\Lambda Q^T$	$A = U\Lambda U^*$
Eigenvalues are real	Eigenvalues are real
Eigenvectors are orthogonal	Eigenvectors are orthogonal ($x^*y=0$)

The Spectral Theorem has the same spirit in both settings:

- Real symmetric matrices use orthogonal diagonalization.
- Hermitian matrices use unitary diagonalization.

Hermitian	Unitary
$A = A^*$	$U^*U = I$

Eigenvalues are real	Eigenvalues have magnitude (1)
Represents self-adjoint transformations	Represents length-preserving transformations
$A = V\Lambda V^*$	$U = V\Lambda V^*$

Property	Unitary Matrix
Definition	$U^*U = I$
Inverse	$U^{-1} = U^*$
Columns	Orthonormal
Preserves length	$\ Ux\ = \ x\ $
Distinct eigenvectors	Orthogonal
Diagonalization	$U = V\Lambda V^*$

Diagonalize a Hermitian matrix

$$A = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix}$$

Step - 1: Check the matrix is hermitian or not with $A^* = A$.

Step - 2: Finding the eigenvalues

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 2 - \lambda & i \\ -i & 2 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda)^2 - i(-i) = 0$$

$\lambda_1 = 3, \quad \lambda_2 = 1$

Since, $-i(-i) = 1$

$$(2 - \lambda)^2 - 1 = 0$$

$$(2 - \lambda) = \pm 1$$

Step - 3 : Find the eigenvector for value = 3

$$\begin{array}{l} (A - 3I)x = 0 \\ \begin{bmatrix} -1 & i \\ -i & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \\ -x + iy = 0 \\ x = iy \end{array} \quad \left| \quad \begin{array}{l} \text{Choose } y = 1, \text{ then } x = i \\ \boxed{v_1 = \begin{bmatrix} i \\ 1 \end{bmatrix}} \end{array} \right.$$

Step - 4: Find the eigenvector for value = 1

$$\begin{array}{l} (A - I)x = 0 \\ \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \\ x + iy = 0 \\ x = -iy \end{array} \quad \left| \quad \begin{array}{l} \text{Choose } y = 1, \text{ then } x = -i \\ \boxed{v_2 = \begin{bmatrix} -i \\ 1 \end{bmatrix}} \end{array} \right.$$

Step-5: Normalize all eigenvectors

$$\|v_1\| = \sqrt{|i|^2 + |1|^2} = \sqrt{1 + 1} = \sqrt{2} \quad u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$\|v_2\| = \sqrt{2} \quad u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

Step - 6: Check Orthogonality

Use the complex inner product:

$$u_1^* u_2 = \frac{1}{2} [(-i)(-i) + 1(1)] \quad \boxed{u_1 \perp u_2}$$

Step-7: Form the Unitary Matrix

Put the normalized eigenvectors as columns.

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}$$

Step 8: Form the Diagonal Matrix

The eigenvalues go in the same order as the eigenvectors:

$$\Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

Step-9 : Final Answer

Since U is unitary,

$$U^{-1} = U^* \quad \boxed{A = U \Lambda U^*}$$